

Hassan Nadeem

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Education

Ph.D. in Bioengineering

University of Illinois Urbana-Champaign (UIUC)

Expected completion: Spring 2027

GPA: 4.00/4.00, Phi Kappa Phi Honor Society

Jan 2023 – Present

M.S. in Nuclear Engineering

Pakistan Institute of Engineering and Applied Sciences (PIEAS), Pakistan

Nov 2015 – Nov 2017

B.S. in Mathematics

National University of Sciences and Technology (NUST), Pakistan

Sept 2011 – Aug 2015

Research Experience

Ph.D. Researcher

University of Illinois Urbana-Champaign (UIUC) | Advisor: Prof. Diwakar Shukla

Jan 2023 – Present

Conformational Ensembles as Determinants of Protein Function

- Developed machine-learning and molecular-simulation approaches to characterize protein conformational ensembles and investigate sequence–structure–dynamics–function relationships.
- Developed **Pi-Ensemble**, a sequence-guided machine learning framework combining inverse folding and protein structure prediction to generate intermediate conformations between functional protein states.
- Developed **EASE** and co-developed **AdaptivePy**, computational frameworks for adaptive molecular simulations that use policy ranking and machine-learning-based sampling strategies to improve conformational exploration and convergence.
- Developed sequence-conditioned generative models for peptide conformational ensembles using molecular-simulation datasets to learn structural distributions across sequence space.
- Developed machine-learning analyses for large-scale experimental peptide sequence–function datasets to characterize enzyme substrate specificity and quantify sequence-level predictive signal.
- Designed and executed GPU-accelerated molecular simulation and machine-learning workflows on HPC systems, including automated simulation, data processing, model training, and analysis pipelines.
- Collaborated with experimental scientists to integrate molecular simulation and machine learning with peptide screening and single-molecule measurements, linking computational predictions to experimentally observed function.
- Research output: **7 publications/preprints, including 5 first/co-first-author papers, and 4 conference presentations/posters, including 3 as first author.**

M.S. Researcher

Pakistan Institute of Engineering and Applied Sciences (PIEAS), Pakistan | Advisor: Prof. Sikander Majid Mirza

Nov 2015 – Nov 2017

Multi-Objective Optimization of Core Loading Patterns for Light-Water Nuclear Reactors

- Developed multi-objective optimization methods for nuclear reactor core loading patterns while balancing competing operational, performance, and safety constraints.
- Applied numerical optimization, reactor-physics calculations, and scientific computing to evaluate candidate loading patterns and support engineering decision-making.

Open-Source Software

Pi-Ensemble

Modular Python software with configurable structure-prediction, inverse-folding, and interpolation backends; includes reproducible workflows, visualization utilities, testing, and containerized deployment. github.com/ShuklaGroup/Pi-Ensemble

AdaptivePy

Installable Python package with a unified API for adaptive simulation, configurable sampling policies, clustering and seed selection, CLI/YAML workflows, documentation, testing, and reproducible run metadata. PyPI: `pip install adaptivepy-sampling`. github.com/ShuklaGroup/AdaptivePy

EASE

Reusable software organized around simulation, policy evaluation, ranking, and benchmarking components for comparing adaptive-sampling strategies. github.com/ShuklaGroup/EASE

Industrial Experience

Senior Scientist

Pakistan Atomic Energy Commission

Dec 2019 – Dec 2021

- Developed and implemented computational optimization algorithms for pressurized light-water reactor core loading, improving fuel-cycle design and operational efficiency.
- Served as a Quality Assurance Engineer, conducting internal audits, coordinating cross-functional corrective actions, and

supporting compliance with regulatory and organizational standards.

- Led radiation protection and nuclear safety training for approximately 300 officers and technicians annually.
- Coordinated technical workshops, regulatory missions, and peer reviews involving national and international organizations, including PNRA, IAEA, and WANO.

Junior Scientist

Nov 2017 – Dec 2019

Pakistan Atomic Energy Commission

- Delivered technical instruction in mathematics, reactor fundamentals, radiation protection, and nuclear safety to engineers and technical personnel.
- Conducted radiation protection training for nuclear power plant personnel in support of operational and regulatory safety requirements.
- Completed cross-functional technical rotations in reactor operations, nuclear-island maintenance, and health physics, developing practical experience with complex safety-critical engineering systems.

Skills / Software

Machine Learning	PyTorch, scikit-learn; generative modeling, representation learning, statistical learning
Programming & Scientific Computing	Python, C++, MATLAB, Shell, Git, Hugging Face, Linux/Unix, HPC
Molecular Simulation	OpenMM, LAMMPS, AMBER, GROMACS, NAMD, MDTraj; adaptive sampling, Markov state models
Molecular Visualization	VMD, ChimeraX, PyMOL

Publications & Preprints

1. **Nadeem, H.[†]**, Kleiman, D. E.[†], Zhou, Y., Leakey, A. D. B., & Shukla, D. (2026). *Pi-Ensemble: Sequence-guided generation of interpolated protein conformational ensembles*. bioRxiv. doi:10.64898/2026.08.12.744498.
2. **Nadeem, H.[†]**, Kleiman, D. E.[†], & Shukla, D. (2026). *AdaptivePy: A unified Python framework for adaptive sampling in molecular dynamics*. bioRxiv. doi:10.64898/2026.08.04.742868.
3. Ouyang, Y.[†], **Nadeem, H.[†]**, Goto, Y., Shukla, D., & van der Donk, W. A. (2026). *Mapping the sequence preference of the generalist class II lanthipeptide synthetase ProcM by mRNA display*. bioRxiv. doi:10.64898/2026.08.19.745792.
4. Samajdar, R.[†], **Nadeem, H.[†]**, Moghe, N., Shukla, D., & Schroeder, C. M. (2026). *Solvent environment influences molecular conformation and electron transport in peptides*. *J. Phys. Chem. Lett.*, 17, 6004–6013. doi:10.1021/acs.jpclett.6c01257.
5. **Nadeem, H.**, & Shukla, D. (2025). *Ensemble adaptive sampling scheme: Identifying an optimal sampling strategy via policy ranking*. *J. Chem. Theory Comput.*, 21, 4626–4639. doi:10.1021/acs.jctc.4c01488.
6. Zhao, C., **Nadeem, H.**, & Shukla, D. (2025). *Structural basis for negative regulation of ABA signaling by ROP11 GTPase*. *J. Chem. Inf. Model.* doi:10.1021/acs.jcim.5c02002.
7. Kleiman, D. E., **Nadeem, H.**, & Shukla, D. (2023). *Adaptive sampling methods for molecular dynamics in the era of machine learning*. *J. Phys. Chem. B*, 127, 10669–10681. doi:10.1021/acs.jpcb.3c04843.

[†]Equal contribution.

Conference Presentations

- **Nadeem, H.** et al. *Pi-Ensemble: Sequence-guided generation of interpolated protein conformational ensembles*. ACS Fall 2026, Chicago, IL. **Oral Presentation**.
- **Nadeem, H.** & Shukla, D. *Ensemble Adaptive Sampling Scheme: Identifying an Optimal Sampling Strategy via Policy Ranking*. ACS Spring 2026, Atlanta, GA. **Oral Presentation**.
- **Nadeem, H.** et al. *Towards Predictive Models of Charge Transport in Peptides*. U.S. Army Center for Synthetic Biology 3rd Annual Review, 2026. **Poster Presentation**.
- Samajdar, R., **Nadeem, H.**, et al. *Active Learning to Understand and Predict the Electronic Behavior of Peptides*. AIChE Annual Meeting, 2024.

Teaching & Mentoring

Teaching Assistant, *Computational Mathematics for Machine Learning and Imaging* (2024) and *Linear Algebra for Biomedical Data Science* (2025), University of Illinois Urbana-Champaign (UIUC).

Selected Honors

Mavis Future Faculty Fellowship, UIUC (2024) | **Excellence in Performance Award**, PAEC (2019) | **Excellence in Research Award**, PIEAS (2017) | **President's Gold Medal**, NUST (2015)